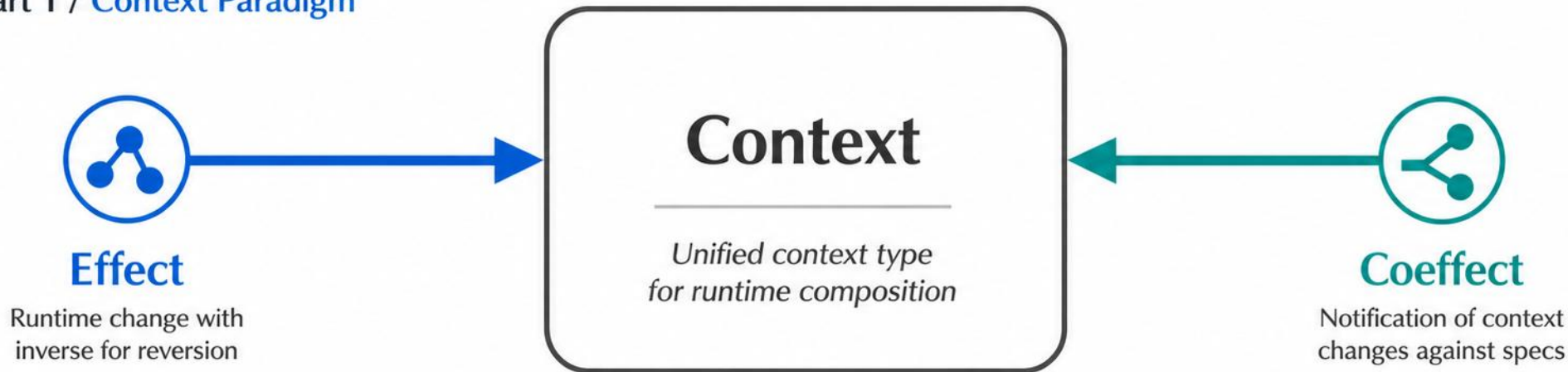


动态组合的新基础

Spatiotemporal Composability

Part 1 / Context Paradigm



A Programming Paradigm for Spatiotemporal Composability

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¹Peking University ²DeepSeek-AI

Abstract

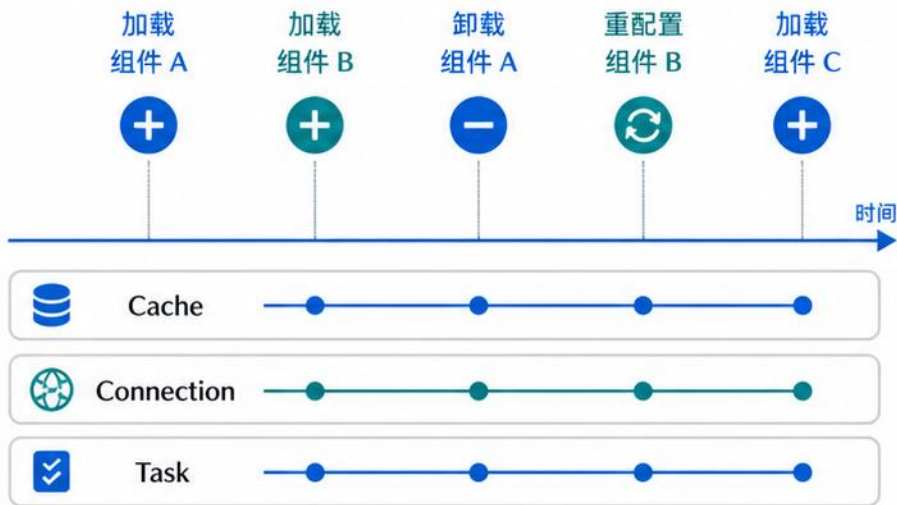
Modern software—from plugin systems to self-evolving agent harnesses—increasingly requires *dynamic composition*, yet its formal foundations remain underdeveloped. We identify two orthogonal dimensions of the problem: *temporal composability*, the ability to completely revert a component's side effects upon removal, and *spatial composability*, the ability to declare and reactively manage inter-component dependencies. We address the two dimensions by lifting classical effect and coeffect concepts to runtime mechanisms. In particular, we formalize *revertible effects* and *reactive coeffects*, unify them into a single *context type*, and give a calculus of dynamic composition whose metatheory carries spatiotemporal composability from a single component to a whole system of interleaved components.

运行时组件变化；重启丢状态；粒度不匹配

Mismatch: Fine-Grained Runtime Changes vs. Coarse Restart Mechanisms

细粒度：运行时组件级变化

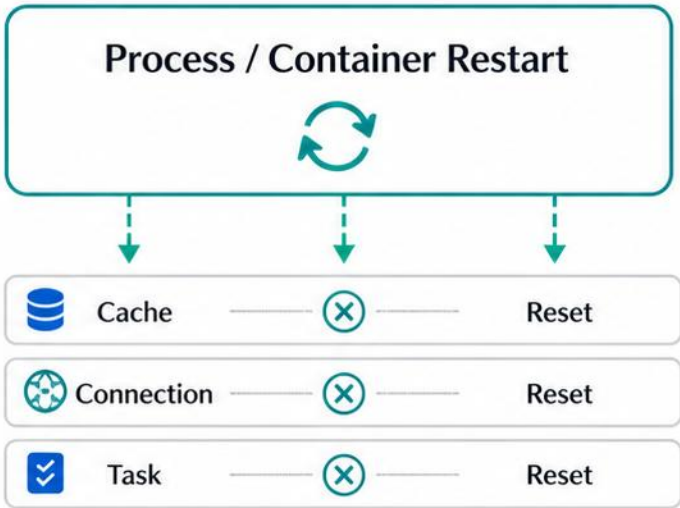
组件可被加载、卸载、重配置，系统状态持续



组件级变化对状态的影响可局部、可恢复

粗粒度：进程/容器级重启

一个组件故障或变更，触发整体重启



重启导致所有状态丢失并重建，代价高昂



粒度不匹配

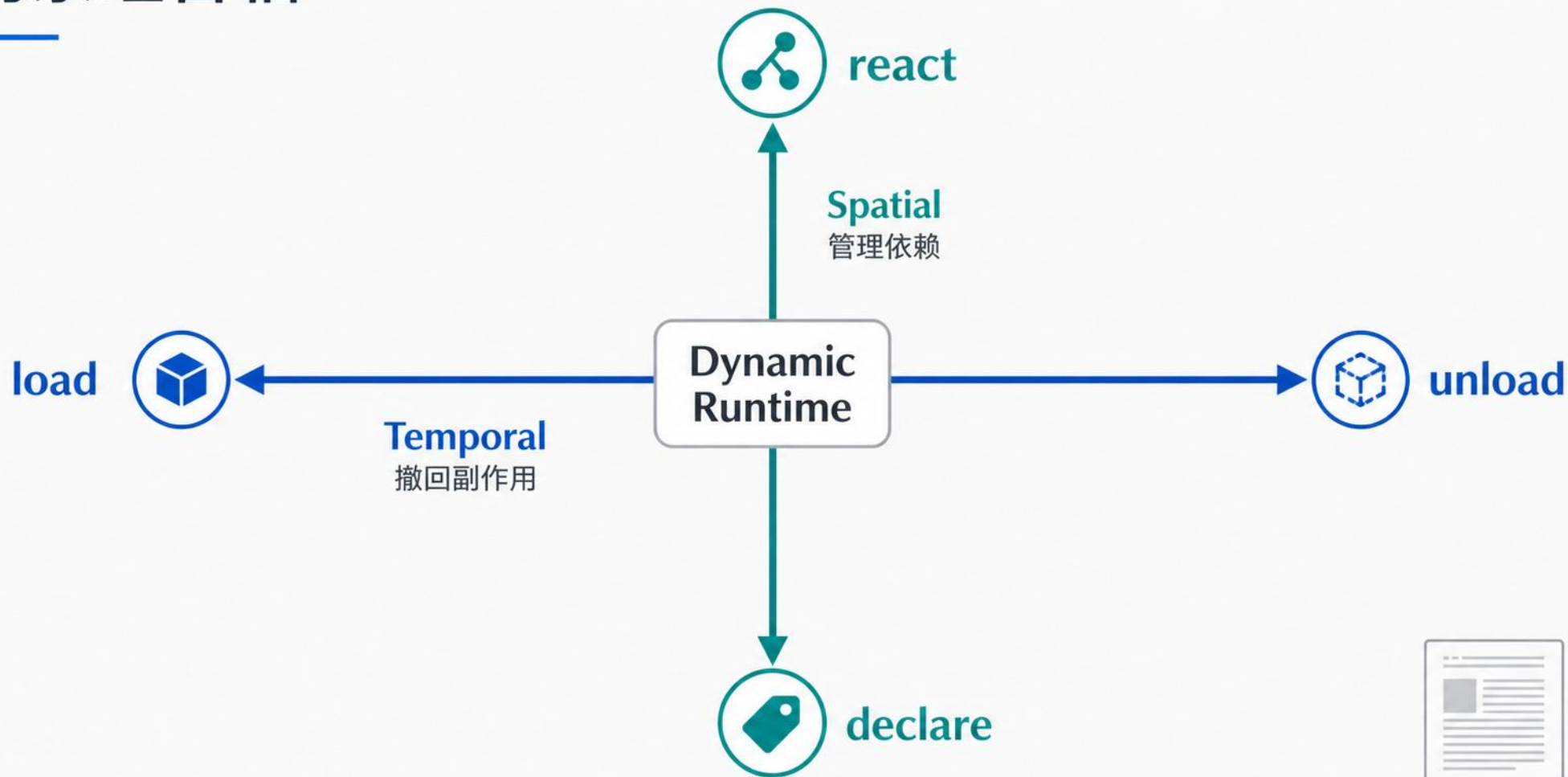
细粒度变化无法由粗粒度重启精确表达，导致不必要的代价与状态丢失

证据

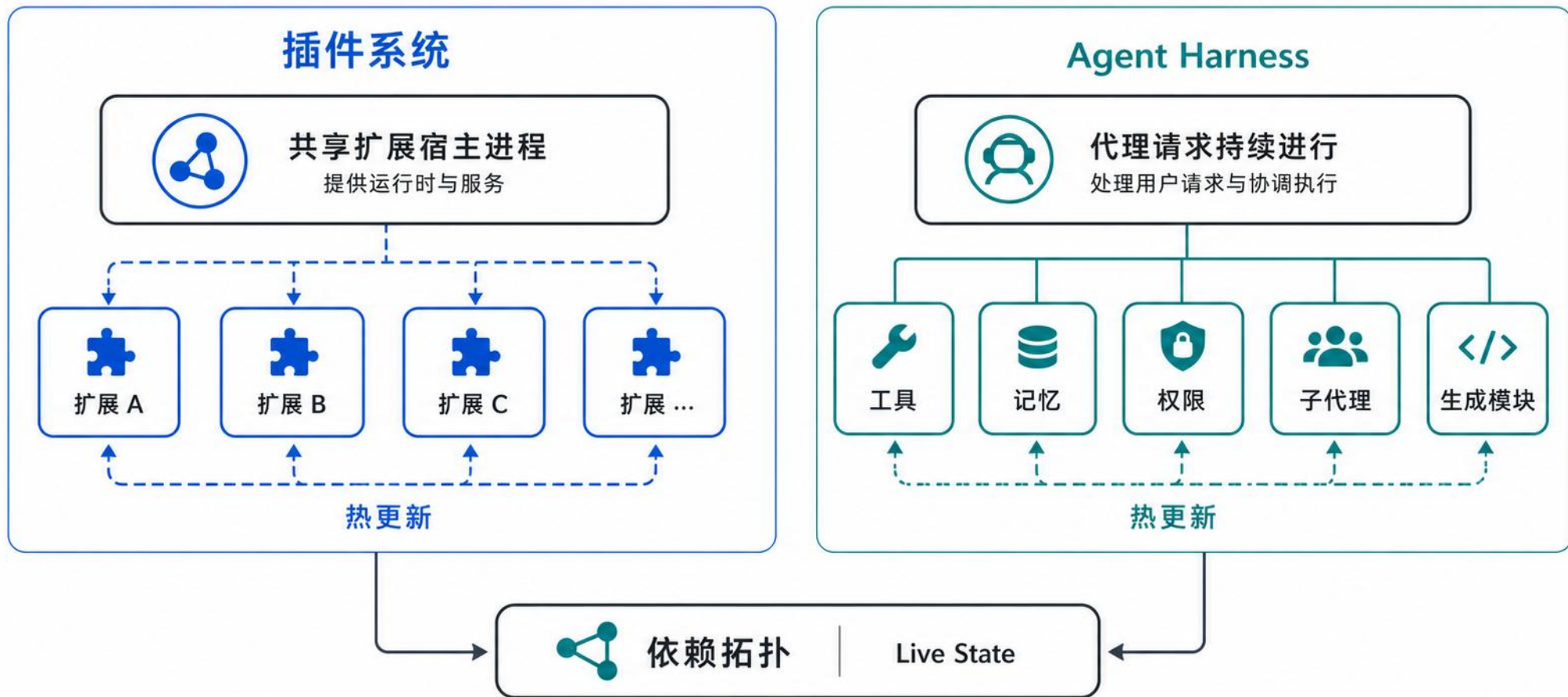
引言：插件架构与自演化智能体需要在运行时安全地增删功能，但当前实践依赖粗粒度机制（重启或丢弃运行时状态）。 [1]

现状：实践中通过重启进程和容器编排来处理粗粒度问题，但带来状态丢失、可用性中断与资源冗余等显著代价。 [3]

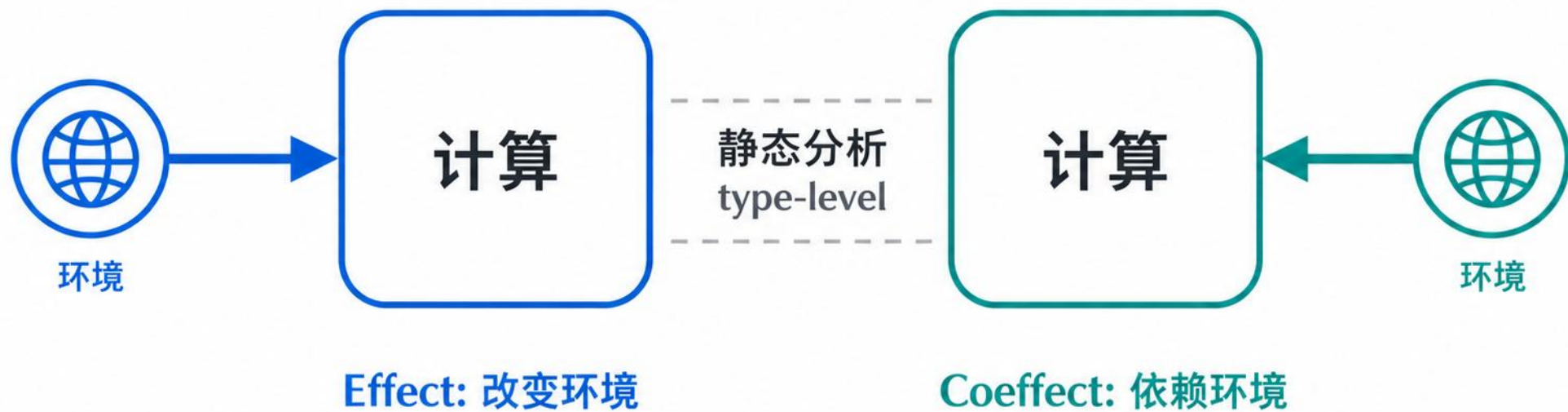
两条组合轴



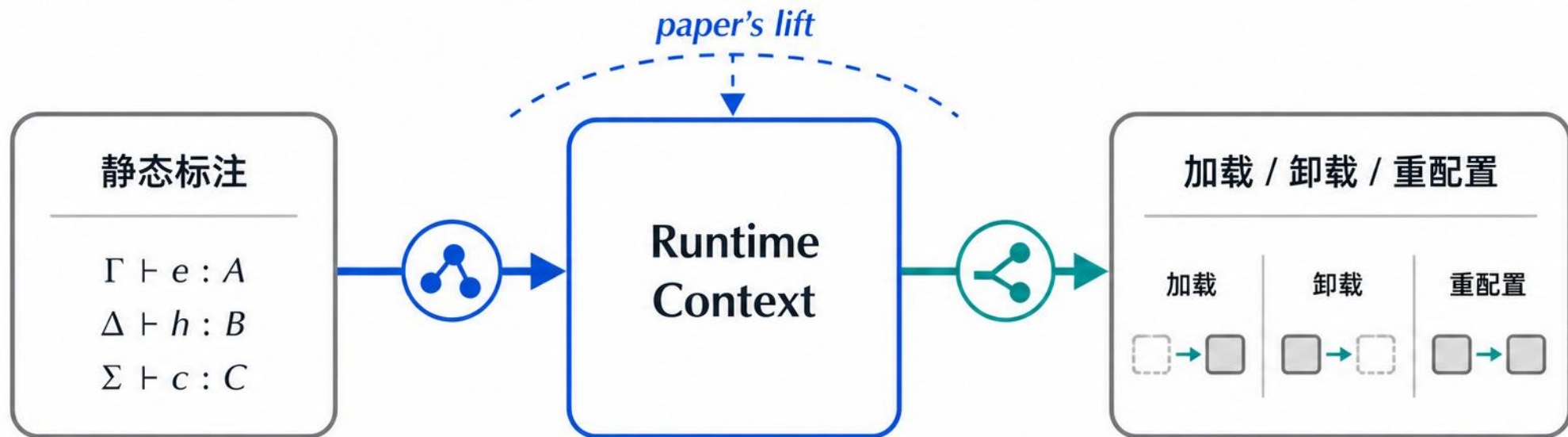
两个现实入口



静态理论的两根柱子

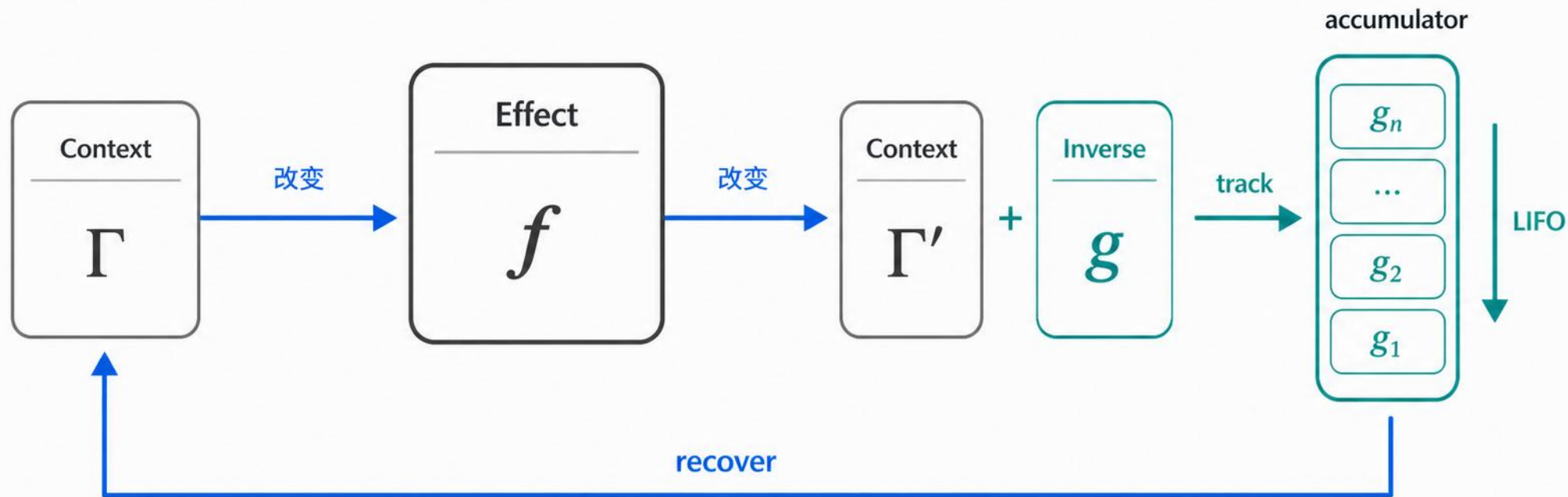


从静态到运行时



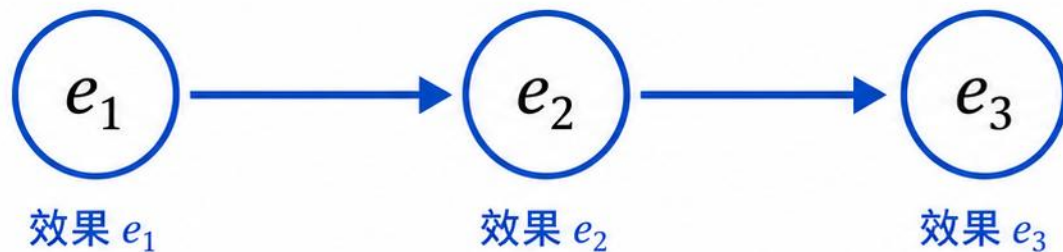
Reversible Effects

改变 + 逆操作

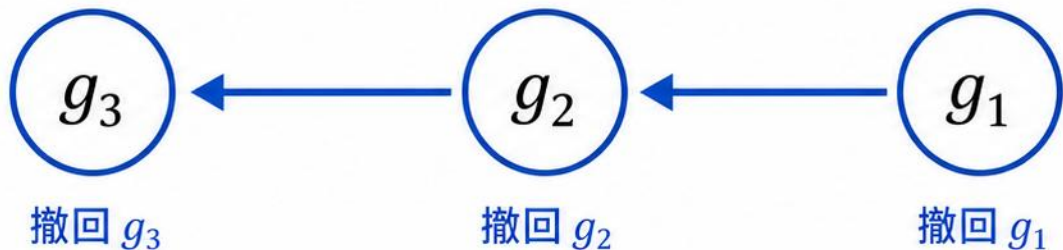


组合为什么还能撤回

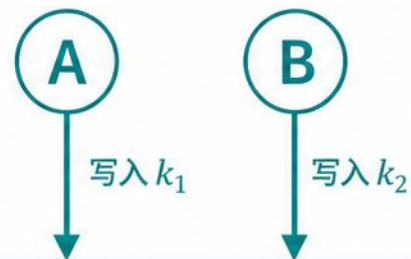
正向组合



反向恢复



独立 key



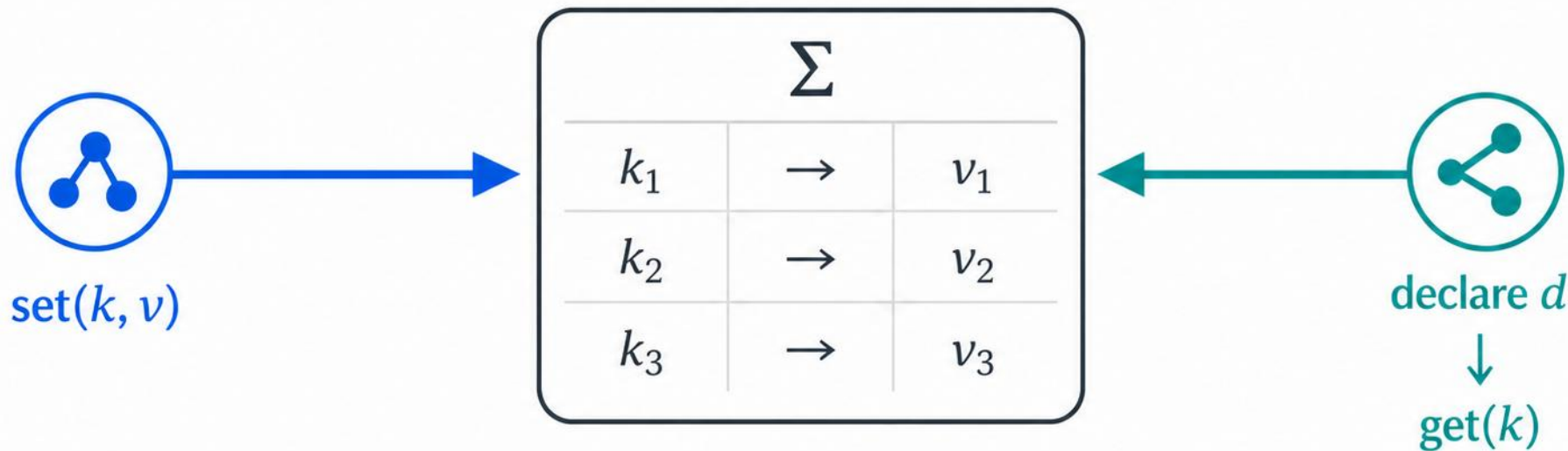
Coeffect 表

Coeffect 表	
k_1	由 A 写入
k_2	由 B 写入

交错也可撤回

Reactive Coeffects

声明依赖



satisfied?



activating



deactivating

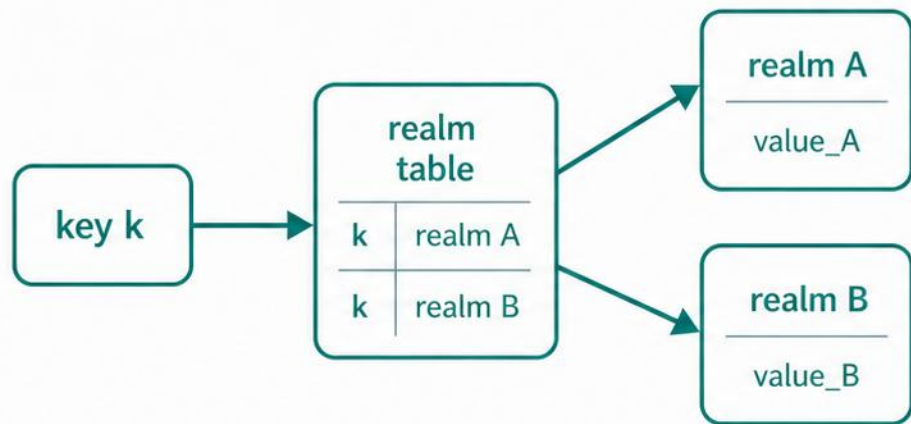


neutral

隔离与拦截

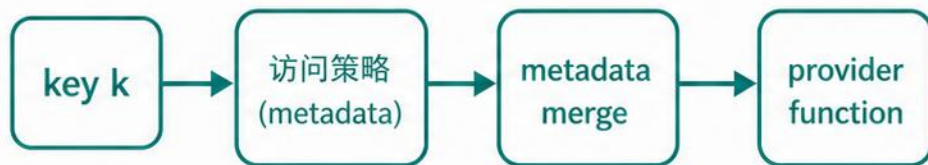
isolate

realm



intercept

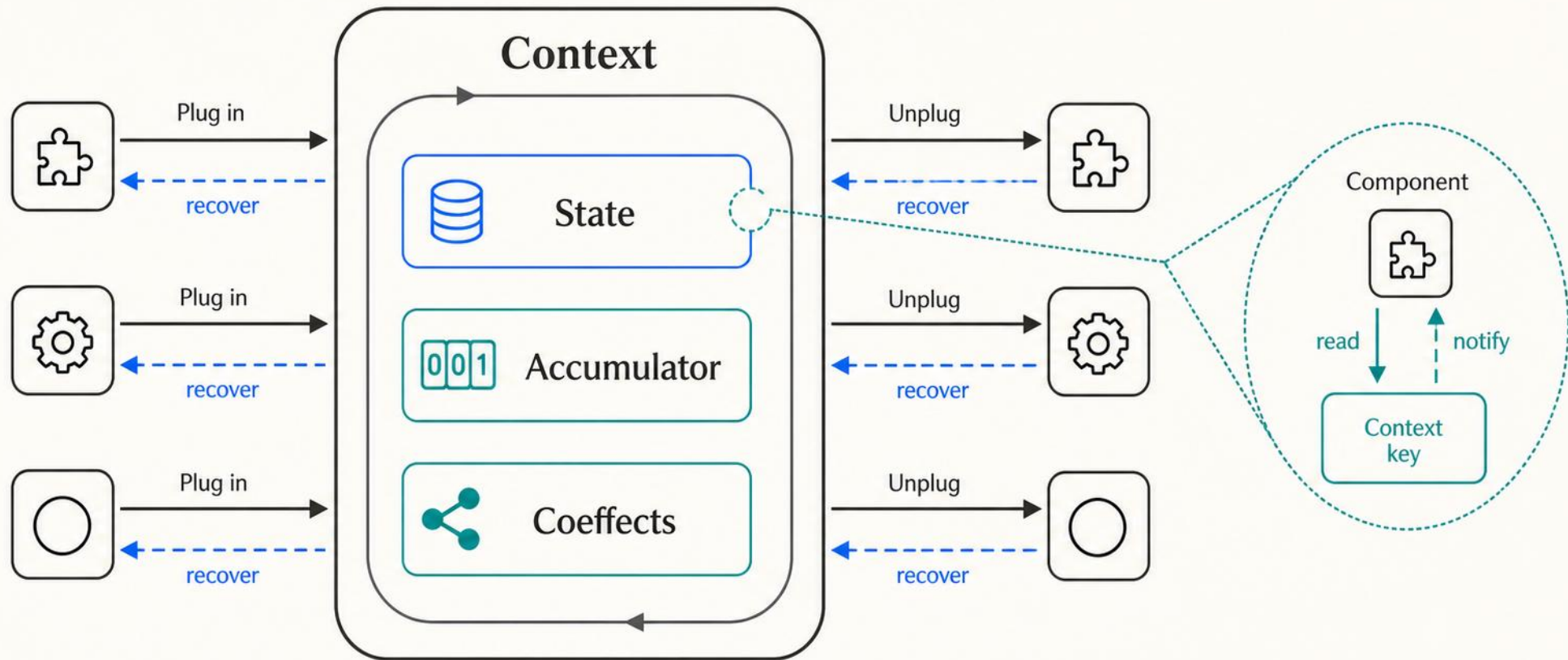
metadata



相同的 key k，受控的解析结果

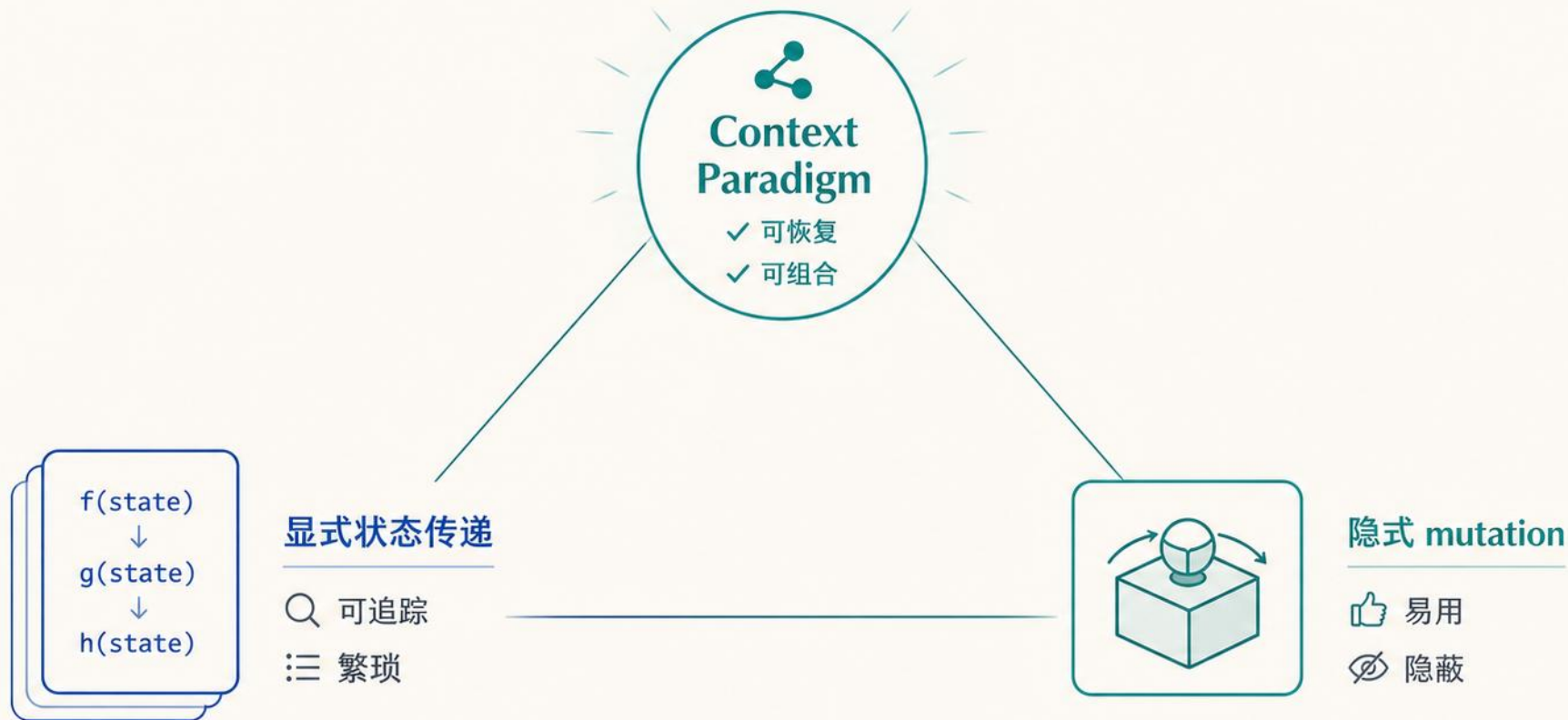
Context Paradigm

统一 Context：把状态、累计器、反应式 coeffect 放进同一个运行时语境



范式的位置

Part 1 小结：在显式状态传递和隐式可变状态之间，寻找可恢复、可追踪的 Context



Part 2: Lifecycle